Information Theory

Test 1, version 1

If the last digit of your student number is odd, then do question 7, but not question 8. If it is even, then do question 8, but not question 7.

This test is worth 6 points + 1 bonus point.

0.5 points

1. State the definitions of ensemble, expected value of a random variable, and entropy of a random variable. State the Markov and Chebyshev inequalities.

0.5

2. Let $x_1, \ldots, x_n$ be the outcome of tossing a balanced 7-sided die $n = 400$ times. I.e., $\Pr[x_i = k] = 1/7$ for every $k \in \{1, \ldots, 7\}$, and the tosses are independent.[1]

   Let $X = \sum_{i=1}^{n} x_i$. Compute the mean and standard deviation of X. Show that

   $$\Pr[X \geq 440] \leq \frac{1}{4}.$$

0.5

3. State the following definitions: symbol code over a set A; average length $L(C, P)$ of a symbol code C under a distribution P over A; prefix-free code; uniquely decodable code.

1

4. Consider the following distribution $P : A \to [0, 1]$:

A	a_1	a_2	a_3	a_4	a_5	a_6	a_7
$P(a)$	1/4	1/4	1/8	1/8	1/8	1/16	1/16

   Compute the entropy of a random variable distributed according to P, obtain the Huffman code associated with it, and compute the average length of this code under P.

1.2

5. State the definition of relative entropy. State (without proving) Jensen's inequality. State (without proving) Kraft's inequality. State and prove Gibb's inequality (you may use, without proof, the fact that $\log x$ is concave). Show that if a symbol code $C : A \to \{0, 1\}^+$ is uniquely decodable, and (x, A, P) is an ensemble, then $L(C, P) \geq H(x)$.

0.3

6. State the definition of H_δ.

[1] Yes, you can make a balanced 7-sided die, using a prism with 7 sides, and appending a heptagonal pyramid at each base of the prism, so it always ends up landing on one of the sides of the prism. You now take the outcome as being the side that's facing *down*. :-)

odd last digit, *2*

7. Prove the:

   Theorem 1 (First part of Shannon's theorem)**.** Let $P : A \to [0,1]$ be a distribution, and let $\delta \in (0,1)$ be any constant. For any given natural number n, let $x_1, \ldots, x_n$ be chosen independently according to P. We then have that

   $$\lim_{n\to\infty} \frac{1}{n} H_\delta(x_1, \ldots, x_n) \leq H(x).$$

   You may need auxiliary results about entropy and standard deviation of independent random variables. You should state these results, but you don't need to prove them.

even last digit, *2*

8. Prove the:

   Theorem 2 (Second part of Shannon's theorem)**.** Let $P : A \to [0,1]$ be a distribution, and let $\delta \in (0,1)$ be any constant. For any given natural number n, let $x_1, \ldots, x_n$ be chosen independently according to P. We then have that

   $$\lim_{n\to\infty} \frac{1}{n} H_\delta(x_1, \ldots, x_n) \geq H(x).$$

   You may need auxiliary results about entropy and standard deviation of independent random variables. You should state these results, but you don't need to prove them.

1 bonus point

9. Suppose that there is a large square plantation field where vegetables are growing, and this field has n^2 hygrometers (humidity-monitoring devices) laid out in an $n \times n$ grid.

   At each sampling time, the devices send to a collector one of three messages, saying whether there is low/medium/high humidity in the vicinity of the device.

   On a typical day, at any given time, on any given position in the field, the probability that the humidity is low is $1 - 2p$, and the probability that the humidity is medium or high is p (each).

   At each sampling time, the collector wishes to communicate the outcome of all measurements to a server. You learned a theorem, let's call it Theorem X, that can tell you the smallest possible length of the collector-to-server message. What is Theorem X? In this particular case, what does Theorem X say should be the code length, as a function of n and p?

   What assumption in Theorem X is actually pessimistic, in this particular scenario, i.e., why is it plausible, in this particular scenario, that one could get an even more efficient compression than what is guaranteed by Theorem X?